imports

IOperations
- cmd_vel(msg: TwistStamped)

IEvents
- scan: Seq(real)

IObstacle
- obstacle: Direction

Direction
- left
- right
- front

Header

TurtleBot
- IOperations
- IEvents
- scan: Seq(real)

Twist
- linear: Vector3
- angular: Vector3

Vector3
- x: real
- y: real
- z: real

TwistStamped
- header: Header
- twist: Twist

IMove
- move(lvel: real, avel: real)

default(): Header

move(lvel: real, avel: real)

IOperations

/cmd_vel(TwistStamped(|header = default(), twist = Twist(|linear = Vector3(|x = lvel, y = 0, z = 0|), angular = Vector3(|x = 0, y = 0, z = avel|)|)|))

→ F

Module

scan: Seq(real) — async → scan: Seq(real)

ref TurtleBot

ref ObstacleDetector

obstacle: Direction

obstacle: Direction

ref Driving

Driving

IOperations
IObstacle

obstacle: Direction

obstacle: Direction

ref stm_ref0 = TurtleBot3FSMC

move(lvel: real, avel: real)

obstacle: Direction

TurtleBot3FSMC

IOperations
IMove
lvel: real, avel: real, PI: real
obs: Direction
C
IObstacle

Driving

obstacle?obs/#C

[obs==Direction::right\/obs==Direction::front]
/move(0, avel)

[obs==Direction::left\/obs==Direction::front]
/move(0, -avel)

/#C

Forward
entry move(lvel, 0)

[since(C)>5]/#C

[since(C)>1]/#C

Turning
entry move(0, avel)

[since(C)>=PI/1]

Rotating